Weekly Assignment: Indirect Argument & the Euclidean Algorithm (Epp 4.7, 4.8, & 4.10)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 4.7 #4). Use proof by contradiction to show that for every integer m, $7m+4$ is not divisible by 7.

Problem 2 (Epp 4.7 #7). Carefully formulate the negation of the following statement, then prove the statement by contradiction.

> There is no least positive rational number.

Problem 3 (Epp 4.8 #11). Determine whether the following statement is true or false. If it is true, prove it; if it is false, give a counterexample.

> The difference of any two irrational numbers is irrational.

Problem 4 (Epp 4.10 #18). Use the Euclidean algorithm to compute $\gcd(5859, 1232)$. Make a trace table showing each step: at each iteration, record the values of a, b, the quotient $q = a \operatorname{div} b$, and the remainder $r = a \bmod b$.